2.3 Mean flow past a side-wall bump

Consider the single-layer quasi-geostrophic shallow water model

$$\frac{\partial q}{\partial t} + J(\psi, q) + \beta \frac{\partial \psi}{\partial x} = 0, \qquad q = (\nabla^2 - k_d^2)\psi \tag{6}$$

in a domain unbounded in $y \geq 0$ and periodic in x with period $L \gg 1$. Find the stationary Rossby wave field (ψ'), in the form of a Fourier series, caused by a zonal flow $U > 0$ passing a small bump of shape $y = \eta(x)$ along the southern side wall ($y = 0$), where

$$\eta(x) = \begin{cases} 1, & |x| < 1, \\ 0, & \text{elsewhere.} \end{cases} \tag{7}$$

Using your preferred plotting software, plot ψ' for $L = 200$, $\beta = 10$, and $U = 1, 10$, and consider whether the changes in wave structure with increasing U make sense.

(Hint: You will likely encounter the following concepts at some point in your solution: linearization, dispersion relation, radiation condition, propagating and evanescent waves.)

Solution.

Linearization: Using the streamfunction convention

$$u = -\psi_y, \qquad v = \psi_x,$$

U has streamfunction $\bar{\psi} = -Uy$. Therefore, we write

$$\psi = -Uy + \psi', \qquad q = \bar{q} + q',$$

where

$$\bar{q} = k_d^2 U y, \qquad q' = (\nabla^2 - k_d^2)\psi'.$$

Substituting into equation (6) and neglecting terms quadratic in the perturbation gives

$$q'_t + U q'_x + (\beta + U k_d^2)\psi'_x = 0.$$

Then,

$$U\partial_x\left[(\nabla^2 - k_d^2)\psi'\right] + (\beta + U k_d^2)\psi'_x = 0.$$

Cancellation gives

$$\partial_x \left(U\nabla^2\psi' + \beta\psi'\right) = 0. \tag{S1}$$

Boundary condition: The boundary function $y = \eta(x)$ is a streamline. Choosing it to be zero gives

$$\psi(x, \eta(x)) = 0.$$

Linearizing about $y = 0$, we obtain

$$-U\eta(x) + \psi'(x, 0) = 0,$$

and hence

$$\boxed{\psi'(x, 0) = U\eta(x).} \tag{S2}$$

Fourier expansion: Since the domain is periodic in x with period L, define

$$k_n = \frac{2\pi n}{L}, \qquad \eta(x) = \sum_{n\in\mathbb{Z}} \hat{\eta}_n e^{ik_n x}.$$

For the bump,

$$\hat{\eta}_n = \frac{1}{L}\int_{-1}^{1} e^{-ik_n x}\, dx,$$

so

$$\hat{\eta}_0 = \frac{2}{L}, \qquad \hat{\eta}_n = \frac{2\sin k_n}{Lk_n} \quad (n \neq 0).$$

Expansion gives:

$$\psi'(x,y) = \sum_{n\in\mathbb{Z}} \phi_n(y) e^{ik_n x}.$$

For each $n \neq 0$, equations (S1) and (S2) imply

$$\phi_n'' + \left(\frac{\beta}{U} - k_n^2\right)\phi_n = 0, \qquad \phi_n(0) = U\hat{\eta}_n.$$

Let

$$K = \sqrt{\frac{\beta}{U}}.$$

Propagating modes and the radiation condition: Substituting $\psi' \propto e^{i(kx+ly-\omega t)}$ into the linearized equation gives

$$\omega = Uk - \frac{(\beta + Uk_d^2)k}{k^2 + l^2 + k_d^2},$$

with group velocity

$$c_{g,y} = \frac{\partial\omega}{\partial l} = \frac{2(\beta + Uk_d^2)kl}{(k^2 + l^2 + k_d^2)^2}.$$

For a stationary mode with $0 < |k_n| < K$, the condition $\omega = 0$ gives

$$l = \pm m_n, \qquad m_n = \sqrt{K^2 - k_n^2}.$$

The radiation condition requires energy to propagate toward $y > 0$, so $c_{g,y} > 0$. Since $\beta > 0$ and $U > 0$, this requires $k_n l > 0$ and therefore selects

$$l = \operatorname{sgn}(k_n) m_n.$$

Applying the boundary condition $\phi_n(0) = U\hat{\eta}_n$, we obtain

$$\boxed{\phi_n(y) = U\hat{\eta}_n e^{i\,\operatorname{sgn}(k_n) m_n y}.}$$

Evanescent modes: For $|k_n| > K$, define

$$\alpha_n = \sqrt{k_n^2 - K^2}.$$

The solutions have the form $e^{\pm\alpha_n y}$. Boundedness as $y \to \infty$ excludes the growing solution, so

$$\phi_n(y) = U\hat{\eta}_n e^{-\alpha_n y}.$$

The stationary equation does not determine the x-independent component. We choose no additional zonal-mean current, so this component is constant:

$$\phi_0(y) = U\hat{\eta}_0 = \frac{2U}{L}.$$

Final solution: Combining positive and negative wavenumbers gives the real Fourier-series solution

$$\boxed{\begin{aligned}\psi'(x,y) = \frac{2U}{L} &+ \frac{4U}{L}\sum_{\substack{n\geq 1\\ k_n<K}} \frac{\sin k_n}{k_n}\cos(k_n x + m_n y)\\ &+ \frac{4U}{L}\sum_{\substack{n\geq 1\\ k_n>K}} \frac{\sin k_n}{k_n} e^{-\alpha_n y}\cos(k_n x).\end{aligned}}$$

Here

$$k_n = \frac{2\pi n}{L}, \qquad m_n = \sqrt{\frac{\beta}{U} - k_n^2}, \qquad \alpha_n = \sqrt{k_n^2 - \frac{\beta}{U}}.$$

The first sum contains outgoing propagating waves, while the second contains evanescent waves.

Comparison of $U = 1$ and $U = 10$. For every propagating stationary mode,

$$k_n^2 + m_n^2 = \frac{\beta}{U}.$$

Consequently, the wavelength is

$$\lambda = \frac{2\pi}{\sqrt{k_n^2 + m_n^2}} = 2\pi\sqrt{\frac{U}{\beta}}.$$

With $L = 200$ and $\beta = 10$, we obtain

U	λ	Propagating modes ($n > 0$)
1	1.99	$n = 1, \ldots, 100$
10	6.28	$n = 1, \ldots, 31$

Increasing U from 1 to 10 increases the wavelength by a factor of $\sqrt{10}$, producing a more widely spaced wave pattern. The cutoff wavenumber $K = \sqrt{\beta/U}$ decreases, so fewer Fourier modes can propagate offshore. More of the short-scale components are instead evanescent and remain confined near the coast.

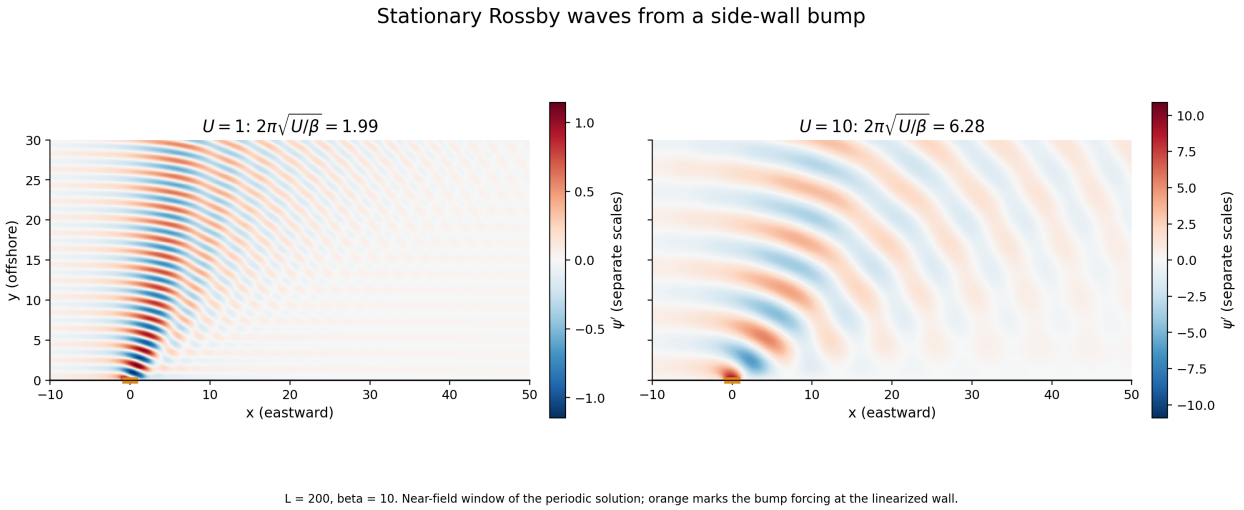


Figure 1: Stationary perturbation streamfunction ψ' for $L = 200$, $\beta = 10$, and $U = 1$ and 10. The figure shows a near-coast window of the periodic solution. Orange segments mark the bump location on the linearized boundary.